



Effects of monetary policy shocks on exchange rate in small open Economies[☆]

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ABSTRACT

This study empirically investigates the effects of monetary policy shocks on exchange rates in four small open economies (United Kingdom, Canada, Sweden, and Australia) by using VAR models in which sign restrictions on impulse responses are imposed to identify monetary policy shocks. The main empirical findings are as follows. (1) A contractionary monetary policy leads to significant exchange rate appreciation. (2) The delay in overshooting is relatively short, at best six months. (3) The deviation from the UIP condition is relatively small, although significant at short horizons in a few cases. To obtain results without strong puzzles, constructing an empirical model that reflects the features of the small open economy and estimates the model for the recent inflation-targeting period during which monetary policy operating procedure does not change substantially are important.

1. Introduction

The effect of monetary policy shocks on exchange rates is a contentious issue. Past studies often found puzzling results, such as “delayed” overshooting. In contrast to predictions of popular theory, such as that proposed by Dornbusch (1976), previous empirical studies often documented that the maximum response of the exchange rate could be observed only with substantial delay. In addition, large and significant deviations from the UIP condition, conditional on monetary policy shocks, are found frequently.

Eichenbaum and Evans (1995) documented that the delay lasts for two to four years by investigating the effects of US monetary policy shocks on the exchange rates of the US dollar against other major currencies with a recursive VAR model (e.g., Sim, 1980). Eichenbaum and Evan (1995) also reported huge, persistent, and significant deviations from the UIP condition, conditional on monetary policy shocks, in the US. By using a recursive VAR model for non-US G7 countries (against the US), Grilli and Roubini (1995) documented even more puzzling responses: contractionary monetary policy shocks result in an exchange rate depreciation (instead of an exchange rate appreciation), which is inconsistent with most theories. This is called the “exchange rate puzzle.”

Many subsequent studies have investigated the issue by using alternative identification methods and/or considering other countries, but the results remain controversial. Some studies, such as Kim and Roubini (2000) (for non-US G7 countries against the US), Kim (2003) (for the US), Cushman and Zha (1997) (for Canada), and Faust and Rogers (2003) (for the US), proposed alternative identification methods and documented that delays in overshooting are relatively short or less certain and that deviations from the

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UIP condition are relatively small or less certain. However, other studies, such as Scholl and Uhlig (2008), Bouakez and Normandin (2010), and Heinlein and Krolzig (2012), reported a long delay in overshooting and significant deviations from the UIP condition for the US. In particular, by using a new identification method that imposes sign restrictions on impulse responses, Scholl and Uhlig (2008) documented that the responses are as puzzling as those reported by Eichenbaum and Evans (1995); the delay lasts for one to three years and the deviation from UIP is huge, persistent, and significant for the US.

This study investigates the effects of monetary policy shocks on exchange rates in small open economies. The studies on small open economies, particularly those focusing on multiple small open economies, are relatively rare. In this study, by analyzing the issues in four small open economies, we draw some general conclusions for small open economies. This study is interested in the following issues, which were documented as puzzles in past studies. First, does a contractionary monetary policy shock lead to a significant exchange rate appreciation? Second, how large is the length of delay in overshooting? Third, is the deviation from the UIP condition significant, conditional on monetary policy shocks?

Scholl and Uhlig (2008) recently studied and documented huge puzzling responses; thus, we use this study as the benchmark. Similar to what Scholl and Uhlig (2008) did, we use the identification method of monetary policy shocks by imposing sign restrictions on impulse responses. The sign restriction method has become very popular at present, but not many previous studies investigated the issues in small open economies by using such a method, which makes the current study more interesting. However, we deviate from Scholl and Uhlig (2008) in two important aspects. First, we incorporate the features of small open economies in our empirical model. Second, we focus on the inflation-targeting period.

We construct the empirical model to account for the features of small open economies to properly reflect the structure of the economies under our investigation, which is different from most previous studies that do not explicitly incorporate the features of small open economies. Cushman and Zha (1997) is one of the exceptions in that emphasized the importance of small open economy modeling for Canada. They documented that the puzzles are weak for Canada by constructing a small open economy model that treats US variables as fully exogenous to Canadian variables. We discuss whether the small open economy modeling is also crucial for small open economies under our investigation.

We consider the period of a homogenous monetary policy regime, which is different from most previous studies that investigated the sample period from the 1970 s. The monetary policy operating procedure for most countries have changed significantly during this period. In particular, many countries adopted the inflation-targeting regime and changed their policy instruments from monetary aggregates to the interest rate in the 1990 s. Such changes in monetary policy operating procedure likely imply substantial changes in monetary policy behavior and its effect on the economy, which may be difficult to capture in a single model estimated over the entire floating exchange-rate regime period. Therefore, we only focus on the inflation-targeting periods. A recent work by Kim et al. (2017) suggested that the effects of US monetary policy shocks on exchange rate can be misleadingly estimated when various monetary policy regimes were pooled and that the delayed overshooting puzzle and the deviation of UIP condition under monetary policy shocks were not significant during the post-Volcker period in the US. We discuss a similar issue for small open economies under our investigations.

We consider four small open economies, namely, the UK, Canada, Sweden, and Australia. We consider advanced economies where a homogenous monetary policy regime (inflation-targeting and floating exchange rate regime) is adopted for a long period (e.g., more than 20 years), and monthly data are readily available. To reflect the features of small open economies, we treat US or world variables as exogenous variables. We also consider variables, such as foreign exchange reserves and US spread, to consider foreign exchange intervention and worldwide risk condition that are likely to affect the exchange rate movements of these countries.

The empirical results show that incorporating small open economy features and considering inflation-targeting period are important in mitigating puzzling responses, such as delayed overshooting and insignificant exchange rate appreciation, found in past studies. During the inflation-targeting period, the exchange rate appreciates significantly with only a small delay in overshooting in these countries.

In Section 2, we explain the empirical method. In Section 3, we report the empirical results from the baseline model. In Section 4, we perform extended experiments to check the robustness of the results. In Section 5, we conclude the study with a summary of the results.

2. Methodology and data

To identify exogenous monetary policy shocks and examine the effects of the identified shocks on the exchange rate, structural vector autoregression (VAR) models with sign restrictions (Uhlig 2005) are used. Previous studies on the effects of monetary policy have frequently used structural VAR models, which are useful in identifying exogenous monetary policy shocks. Several puzzling responses (e.g., liquidity and price puzzles) can be eliminated by imposing proper sign restrictions on impulse responses.¹ The identification strategy with sign restrictions is appealing because puzzling responses are often regarded as failures in identifying proper monetary policy shocks. The methodology of the structural VAR model with sign restrictions was explained in Uhlig (2005) and also briefly in Appendix.

¹ An exogenous monetary expansion (contraction) is supposed to increase (decrease) monetary aggregates and price levels and decrease (increase) interest rates. However, in a model that uses innovations in broad monetary aggregates as monetary policy shocks, both monetary aggregates and interest rates increase. This phenomenon is called the “liquidity puzzle.” On the other hand, in a model that uses innovations in interest rates as monetary policy shocks, both interest rates and price levels increase. This phenomenon is called the “price puzzle.” These puzzles are often regarded as indications that exogenous shocks to monetary policy are not properly identified in the model. Refer to Sims (1992); Christiano, Eichenbaum, and Evans (1999); and Kim (2013) for a survey of the literature.

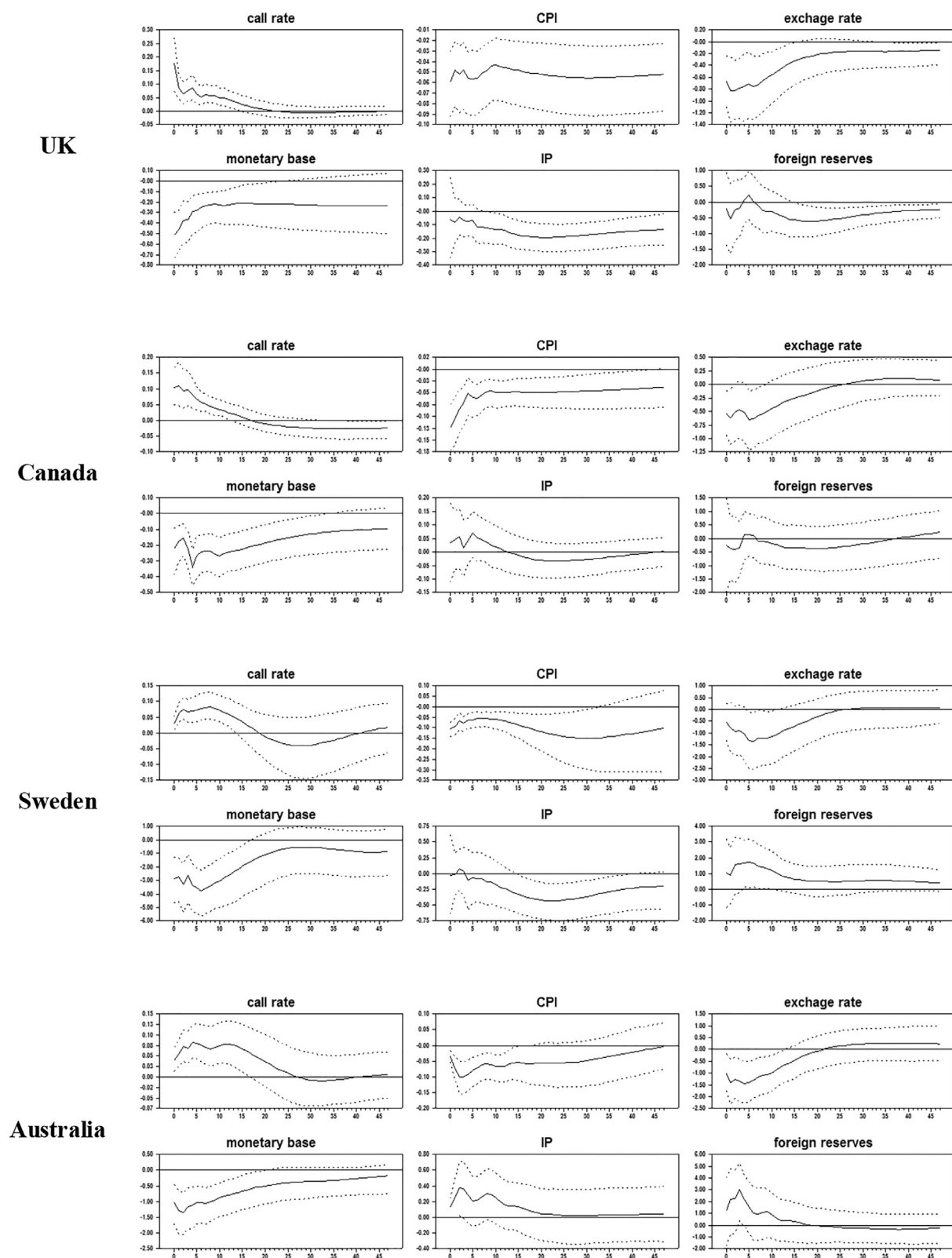


Fig. 1. Impulse responses to contractionary monetary policy shocks.

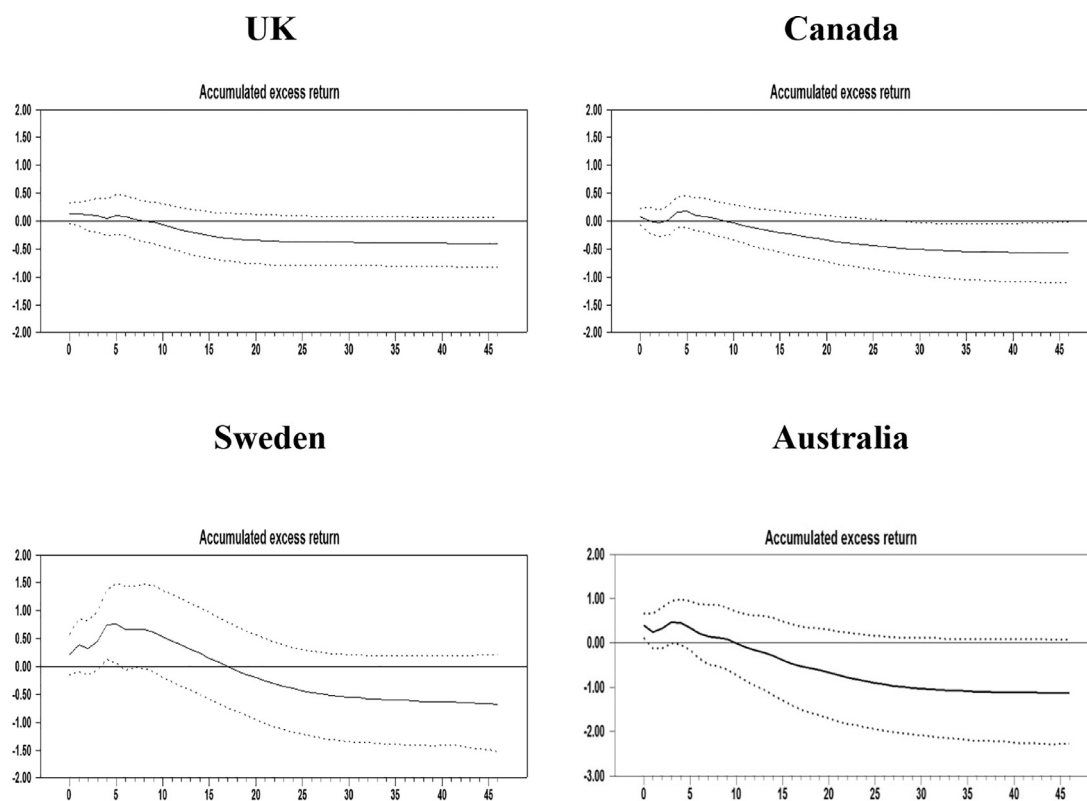


Fig. 2. Accumulated risk premium conditional on monetary policy shocks.

The model developed in this study aims to identify monetary policy shocks in small open economies. In particular, the model is different from that of [Scholl and Uhlig \(2008\)](#) who identified monetary policy shocks for the US, which is regarded as a large open economy. Different from the approach used by [Scholl and Uhlig \(2008\)](#), our method treats the US or world variables as exogenous variables. In addition, some variables, such as foreign exchange reserves, are included to account for foreign exchange intervention in these economies.

Six endogenous variables are included in the baseline VAR model: call rate (CR), monetary base (MB), consumer price index (CPI), industrial production (IP), exchange rate of domestic currency against the US dollar (ERA), and foreign exchange reserves.² The first four variables are key macro/monetary variables that are included to identify monetary policy shocks. The exchange rate is the focus of this study. We also include foreign exchange reserves because the exchange rate may depend critically on foreign exchange rate intervention and omitting foreign exchange reserves may lead to mismeasurement of the effect in these economies.³

Three variables are also included as exogenous variables in the baseline model, namely, the federal funds rate (FFR), IP of the US, and CPI of the US. The FFR is included to control for the US monetary policy, which is likely to affect the exchange rate. Industrial production and the CPI of the US are included because these macro fundamentals are likely important in determining the exchange rate against US dollars in each country. These exogenous variables are also not restricted in terms of their contemporaneous effect on endogenous variables in the model.

The following sign restrictions on impulse responses are imposed to identify contractionary monetary policy shocks: (1) CR increases, (2) MB decreases, and (3) CPI decreases. By imposing these restrictions, liquidity and price puzzles are avoided through construction. Thus, the impulse responses of these basic macro variables to monetary policy shocks are consistent with conventional wisdom on the effects of monetary policy. The sign restrictions are imposed on the impulse responses for the first 12 months after a shock, following [Scholl and Uhlig \(2008\)](#).

Monthly data are used. Details on data definitions and sources are found in the [Appendix](#). Six lags for endogenous variables are assumed. Following [Uhlig \(2005\)](#) and [Scholl and Uhlig \(2008\)](#), neither a constant term nor a time trend is included in the model. Following most previous studies, such as [Eichenbaum and Evans \(1995\)](#), [Kim and Roubini \(2000\)](#), and [Scholl and Uhlig \(2008\)](#), all variables are estimated in levels. Logarithm is taken for IP, CPI, ERA, MB, IP of the US, and CPI of the US. To save the degree of freedom, no lags are assumed for exogenous variables in the baseline model, but certain lags are allowed in the extended models (in [Section 4](#)) to verify the robustness of the results. For each country, we consider the sample period during which inflation-targeting

² M1 is used instead of MB in the UK because M1 is not available.

³ Refer to [Kim \(2005\)](#) who suggested that delayed overshooting can be related to foreign exchange intervention in a small open economy, such as Canada.

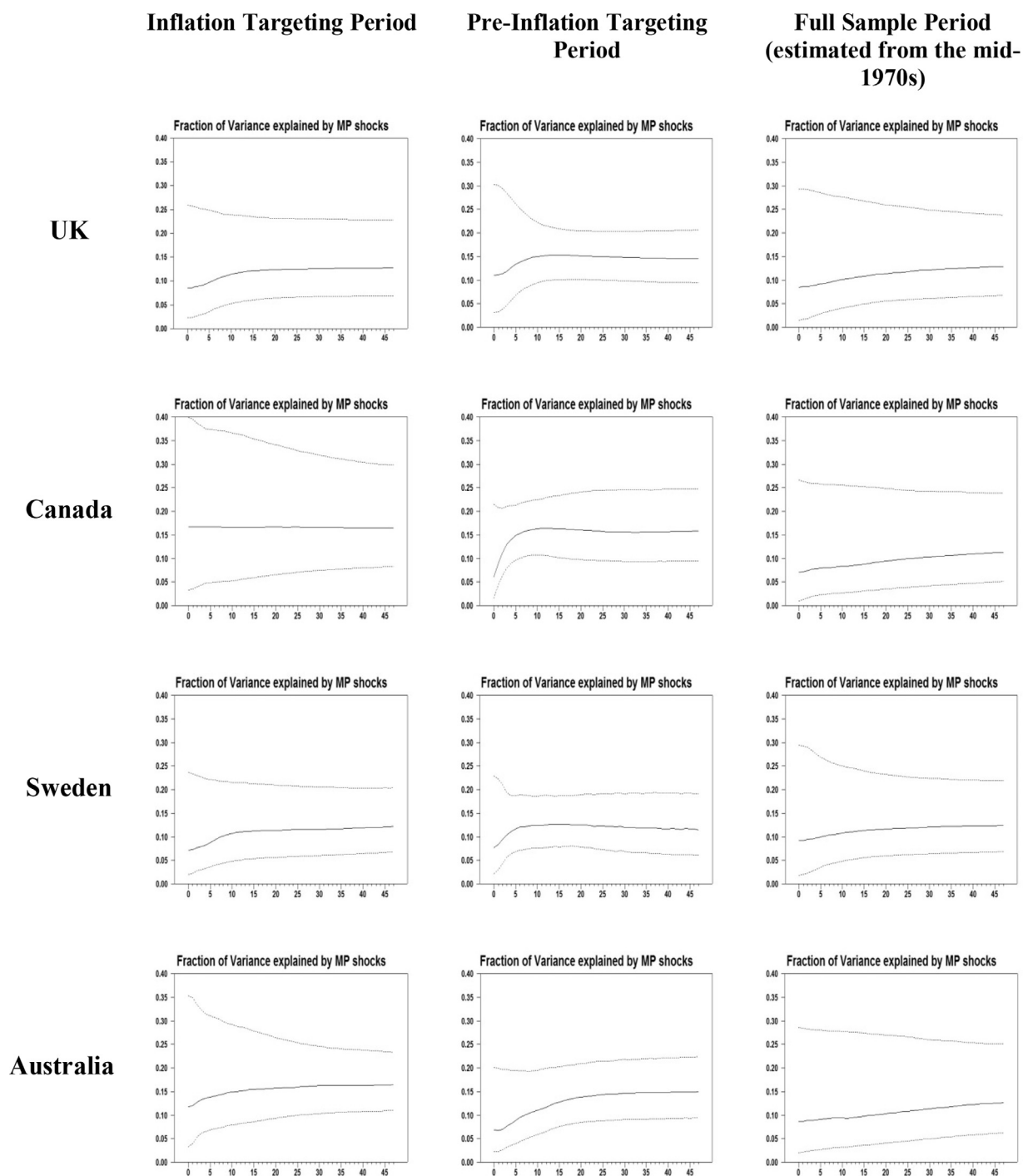


Fig. 3. Forecast error variance decomposition: contribution of monetary policy shocks to exchange rate fluctuations.

period is adopted. The estimation periods are 1992:10–2014:9, 1991:2–2014:9, 1993:1–2014:9, and 1993:3–2014:9 for the UK, Canada, Sweden, and Australia, respectively.

3. Empirical results

3.1. Basic results

Fig. 1 reports the (median) impulse responses to monetary policy shocks with 68% probability bands for each country. The

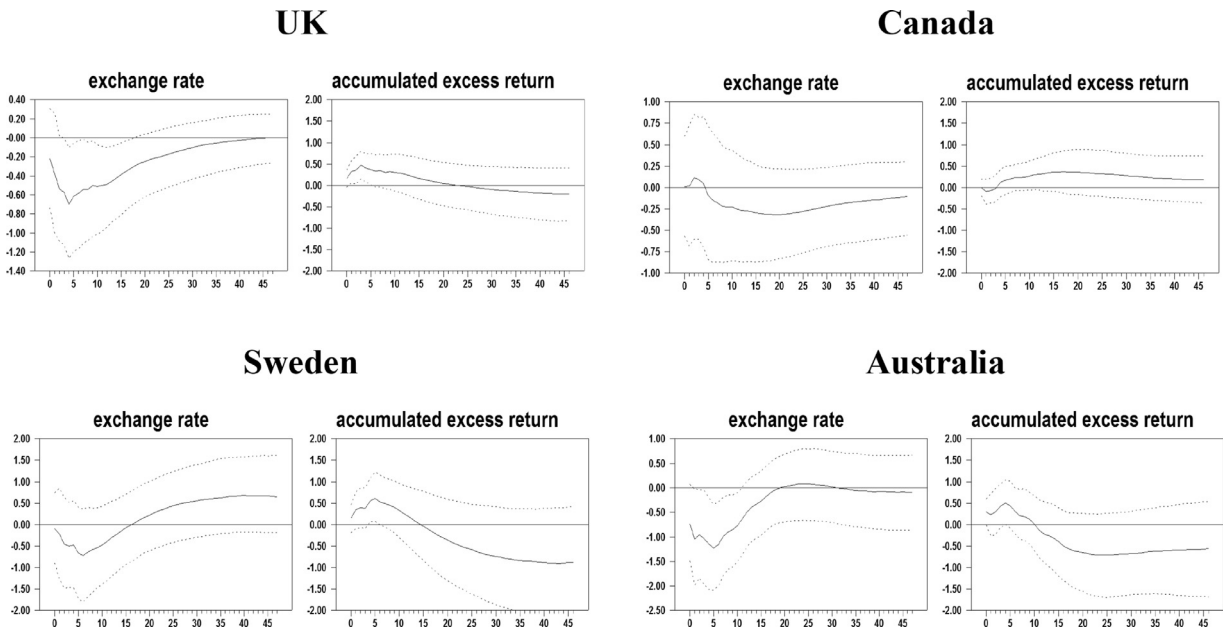


Fig. 4. Model by Scholl and Uhlig (2008).

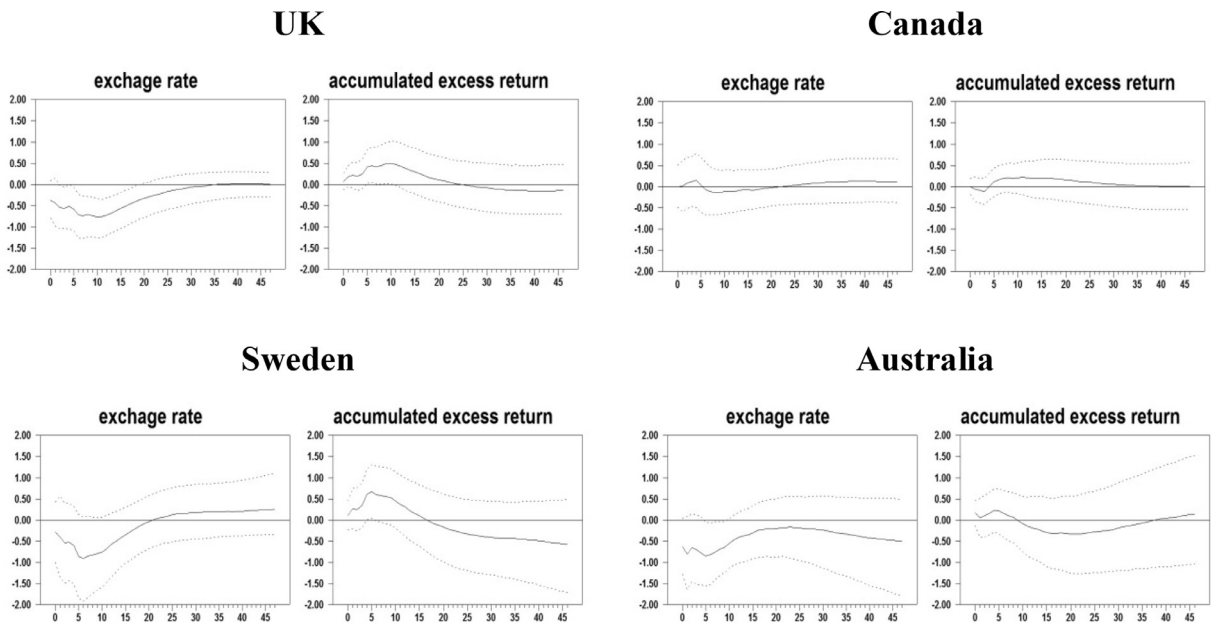
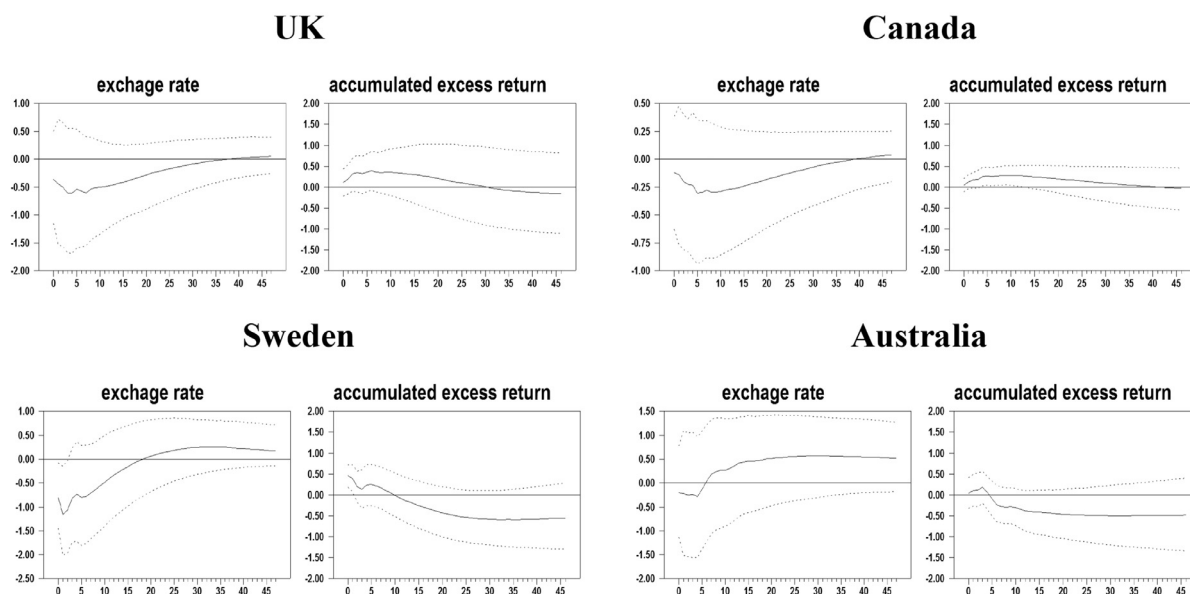


Fig. 5. Dropping exogeneity assumption.

interest rate increases in the short run, the MB decreases, and the CPI declines in all countries. The price and liquidity puzzles are not found by construction because sign restrictions on impulse responses of these variables are imposed. The effects on IP are negative and significant in the UK and Sweden, but error bands are wide and include zero responses in Canada and Australia.

In all countries, the exchange rate appreciates, which is consistent with most theories. Exchange rate appreciation is significant in all countries. The maximum appreciation is observed within six months in all countries. That is, the length of the “delay” in overshooting is extremely small, and almost immediate overshooting is observed. In the UK, the maximum appreciation is observed within one month after the shock, based on the median response. In the three other countries, the maximum appreciation is found within four to six months after the shock, based on the median response. The length of the delay determined in this study is significantly less than that previously reported, such as in Scholl and Uhlig (2008), who observed a delay of 12 to 24 months. The sizes of maximum appreciation are approximately 0.8%, 0.7%, 1.3%, and 1.5% in the UK, Canada, Sweden, and Australia, respectively.

(1) Full Sample Period (estimated from the mid-1970s)



(2) Pre-Inflation Targeting Period

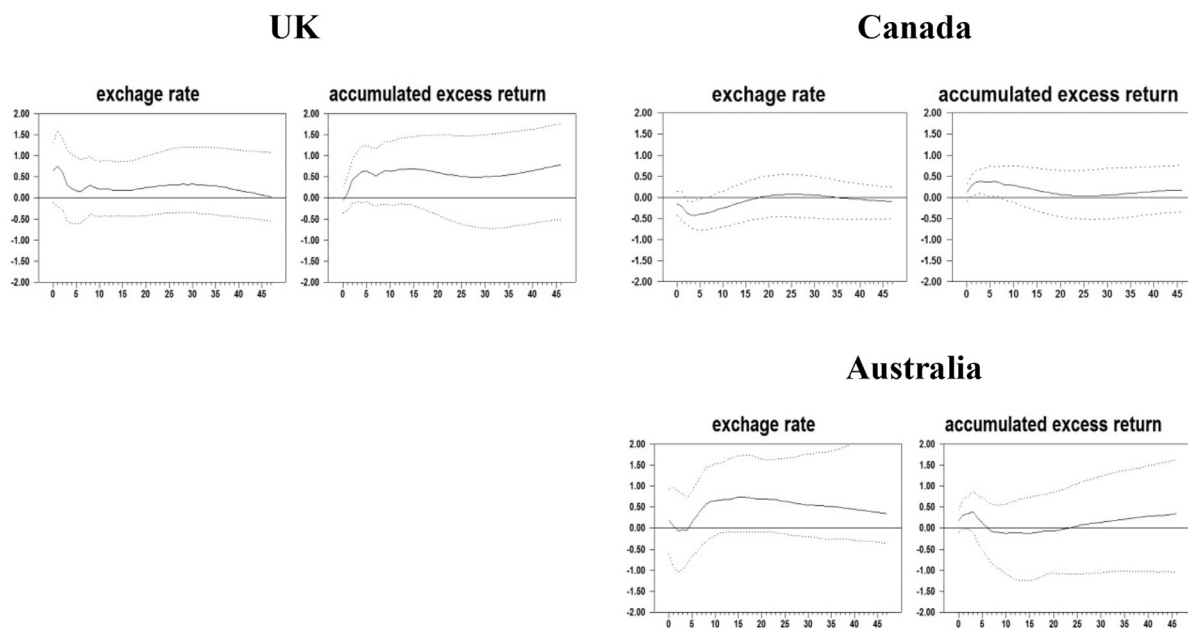


Fig. 6. Estimation for alternative sample periods.

Fig. 1 also shows the impulse responses of foreign exchange reserves. Foreign exchange reserves tend to increase in Sweden and Australia in the short run, which may be interpreted as foreign exchange intervention to stabilize the exchange rate appreciation following monetary contraction. Such foreign exchange intervention may result in delay in overshooting. Kim (2005) provided a related discussion for Canada.

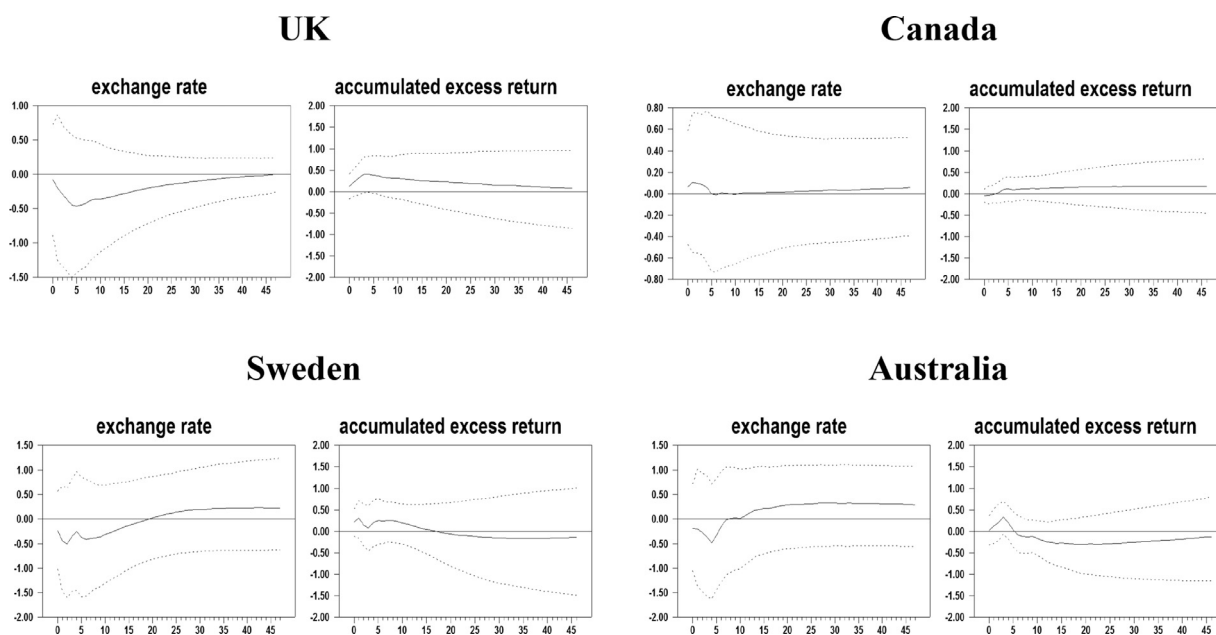


Fig. 7. Model by Scholl and Uhlig (2008) estimated from the mid-1970 s.

Fig. 2 reports the responses of the accumulated risk premium. On the basis of the previous studies, such as Eichenbaum and Evans (1995), the risk premium or the deviation from the UIP condition at t is calculated as follows:

$$\rho_t = (i_t - i_t^*)/12 + S_t - S_{t+1}, \quad (1)$$

where i , i^* , and S are the domestic interest rate, foreign interest rate, and exchange rate, respectively. The risk premium is accumulated over time.

Some cases in which the accumulated risk premium differs significantly from zero, although small, are found. In the UK, it is not significantly different from zero based on such a probability band. In Canada, the accumulated risk premium is significantly different from zero at horizons of over 29–48 months. In Sweden and Australia, it differs significantly from zero within a few short-run horizons.

Fig. 3 reports the contribution of monetary policy shocks on exchange rate fluctuations (the median estimate with 68% probability bands) based on forecast error variance decomposition. In each graph, the x-axis shows the forecast horizons, whereas the y-axis shows the fraction of the forecast error variance explained by monetary policy shocks. On the basis of the median estimate, Canada has the largest contribution, with approximately 17%–18%. The contribution is similar over various horizons. In the other three countries, the contribution tends to increase as the forecast horizon increases. The contribution increases from 9% to 13% in the UK, from approximately 7% to 13% in Sweden, and from 11% to 16% in Australia. Overall, the contribution of monetary policy shocks is not huge but non-negligible.

3.2. Sample periods and identification

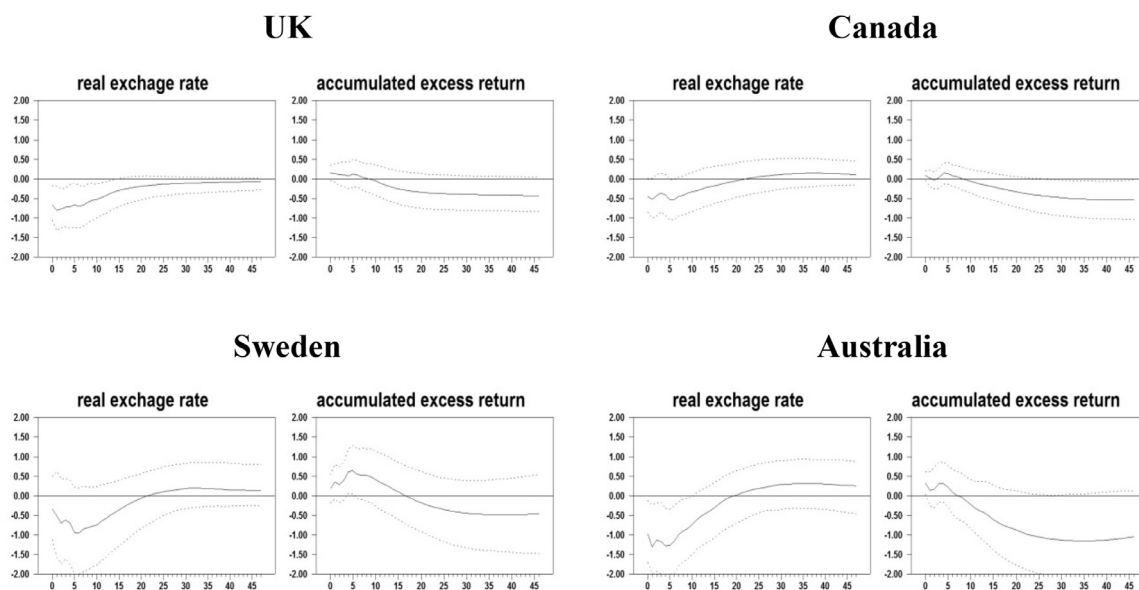
To explain the exchange rate responses without strong puzzles, some extended experiments are performed. First, we intend to confirm that using a small-country identification scheme developed in this study is important. As the benchmark model for comparison, the model developed for the US, which has a large economy, by Scholl and Uhlig (2008) is used. Scholl and Uhlig (2008) included seven variables in the model, namely, CPI, IP, monetary aggregate, interest rate of the home country, interest rate of the foreign country, IP of the foreign country, and exchange rate of a home country against a foreign country, following Eichenbaum and Evans (1995). Eichenbaum and Evans (1995) and Scholl and Uhlig (2008) originally developed a model to investigate the effect for the US, so they treated the US as the home country. In the present study, we treat the UK, Canada, Sweden, and Australia as home countries and the US as the foreign country.⁴ We use the same sign restrictions in the baseline model as those used by Scholl and Uhlig (2008), and we keep the same sign restrictions in the present study.⁵

Fig. 4 presents the results. In Canada, the exchange rate responses are small and insignificant at all horizons. In Canada, the median responses further show that the impact effect is near zero and that the maximum appreciation is found in approximately 20

⁴ Scholl and Uhlig (2008) used nonborrowed reserves of the US as monetary aggregate of home country, but we use monetary base as the monetary aggregate of the home country.

⁵ The sign restrictions are essentially the same as those used in the present paper, namely, (1) the interest rate increases, (2) monetary aggregate decreases, and (3) CPI decreases.

(1) Impulse Responses



(2) Forecast Error Variance Decomposition

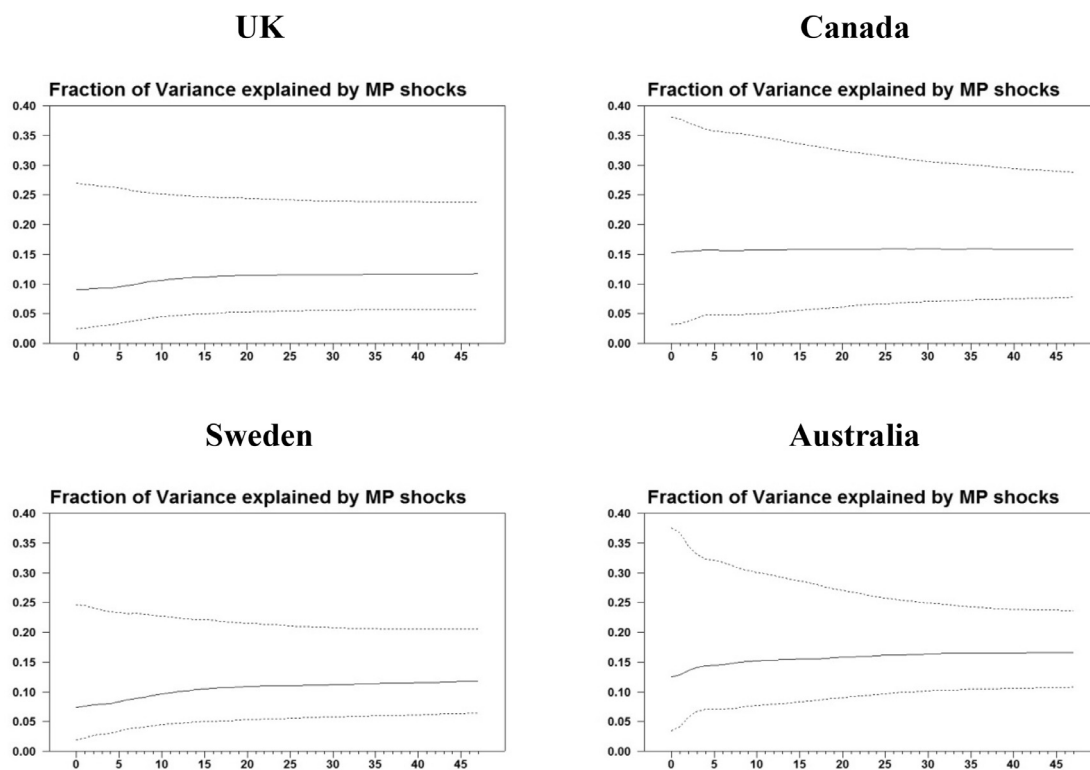
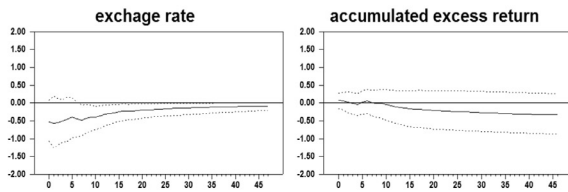


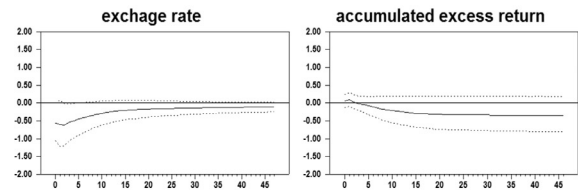
Fig. 8. Models with real exchange rate.

UK

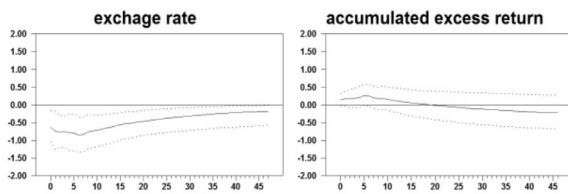
(1) Spread as an Exogenous Variable



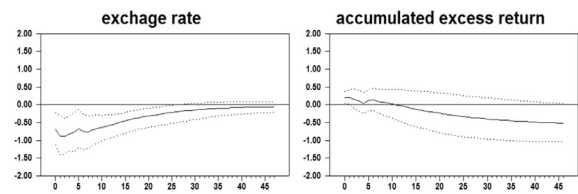
(5) Three Lags for Exogenous Variables



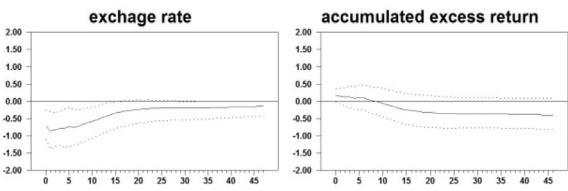
(2) Global Financial Crisis Dummy (08.7–08.12)



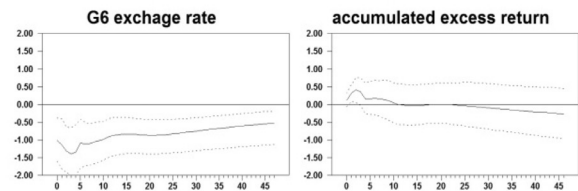
(6) Six Lags for Exogenous Variables



(3) Shadow FFR



(7) G7 Aggregates instead of the US



(4) Non-borrowed Reserves to the Total Reserves as an Exogenous Variable

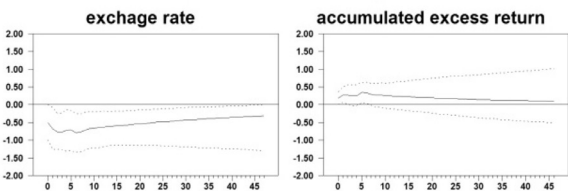


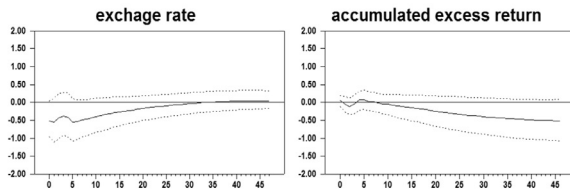
Fig. 9. Robustness.

months after the shock. This result is in line with the argument of [Cushman and Zha \(1997\)](#) where small open economy modeling is found crucial for Canada. For the other three countries, the impact responses are insignificant. In Sweden, responses are not significant at any horizons. The median responses show about six month delays in overshooting in these countries. Immediate overshooting is less clear compared with the results in the baseline model. The accumulated risk premium is different significantly from zero at some initial horizons in the UK, Sweden, and Australia.

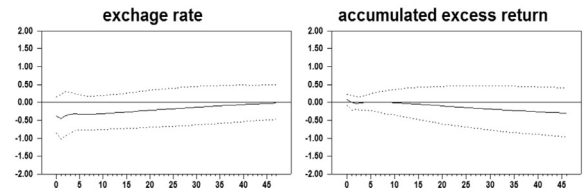
We also try to evaluate how important is the assumption that the US variables are exogenous in the baseline model of the current study because such an assumption is a standard one for small open economics. We construct the model in which all nine variables (six domestic and three US variables) are included as in the baseline model but both domestic and US variables are treated as endogenous variables.

Canada

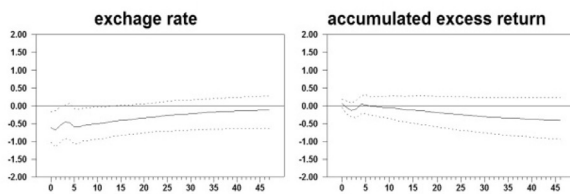
(1) Spread as an Exogenous Variable



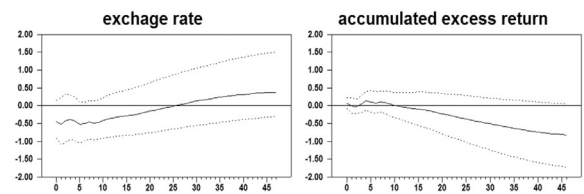
(5) Three Lags for Exogenous Variables



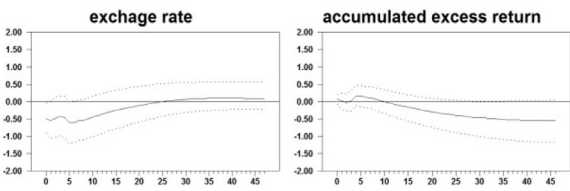
(2) Global Financial Crisis Dummy (08.7–08.12)



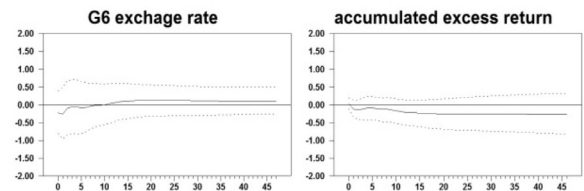
(6) Six Lags for Exogenous Variables



(3) Shadow FFR



(7) G7 Aggregates instead of the US



(4) Non-borrowed Reserves to the Total Reserves as an Exogenous Variable

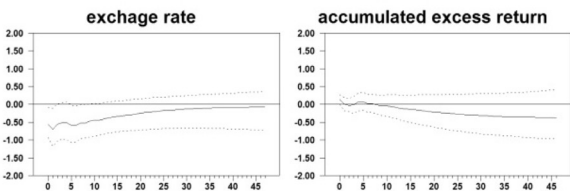


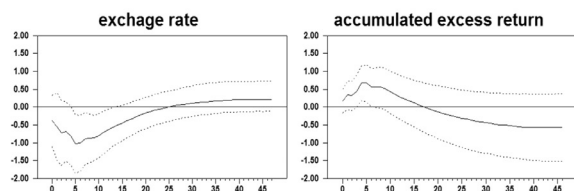
Fig. 9. (continued)

Fig. 5 reports the results. In Canada, the exchange rate responses are small and insignificant at all horizons. For the other three countries, the impact responses are insignificant. In Sweden, responses are not significant at any horizons. The median responses show about six month delays in overshooting in the UK, Sweden, and Australia. To obtain less puzzling exchange rate responses, the assumption (that the US variables are exogenous) seems important for these countries, especially for Canada.

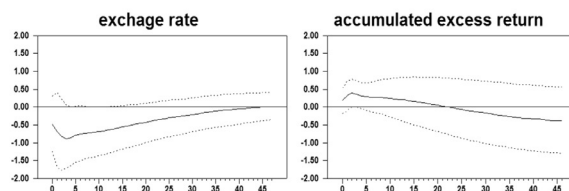
Second, past studies often used the sample period from the mid-1970s when the floating exchange rate regime was adopted. Although the monetary policy operating procedure has changed from time to time, the usual VAR models do not consider such changes. A significant deviation from the UIP condition or a long delay in overshooting found in past studies might be due to such a problem. In this regard, we also estimate the baseline model for the sample period that starts from the mid-1970s (full sample period) and for the sub-sample period that starts from the mid-1970s and stops just before the inflation-targeting period (pre-inflation-targeting period). We compare the results for the inflation-targeting period (inflation-targeting period) with the results for these

Sweden

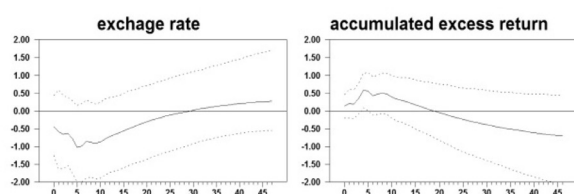
(1) Spread as an Exogenous Variable



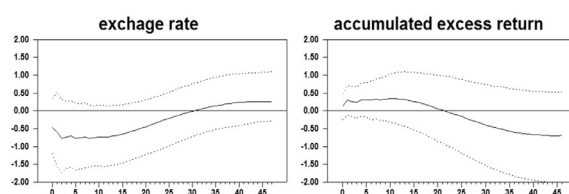
(5) Three Lags for Exogenous Variables



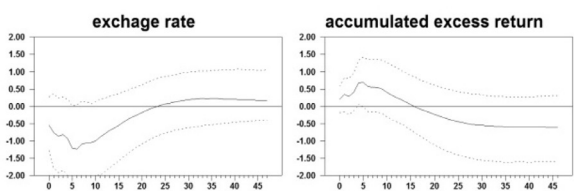
(2) Global Financial Crisis Dummy (08.7–08.12)



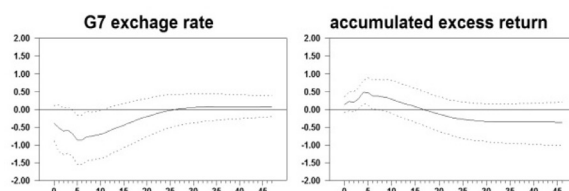
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(4) Non-borrowed Reserves to the Total Reserves as an Exogenous Variable

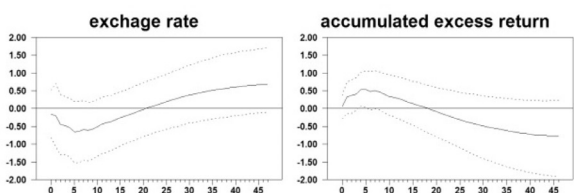


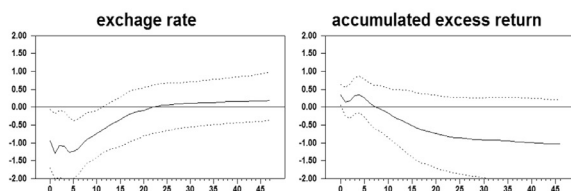
Fig. 9. (continued)

alternative sample periods. The full sample and pre-inflation-targeting periods start from 1975:7 for the UK, Canada, and Australia. For Sweden, the full sample period starts from 1981:1 but the model is not estimated for the pre-inflation-targeting period because available data for the pre-inflation-targeting period are too small.

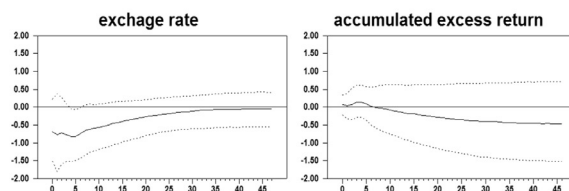
Fig. 6 reports the results when the model is estimated for full sample and pre-inflation-targeting periods. In the UK and Australia, exchange rate responses are insignificant at any horizons for the pre-inflation-targeting period. In addition, the median response suggests that the exchange rate depreciates, showing the exchange rate puzzle. For the full sample period, the exchange rate responses are insignificant but the median response suggests that the exchange rate appreciates, which is consistent with the theory. However, the size of the maximum appreciation is far smaller than that in the inflation-targeting period (0.2% vs. 1.5% in Australia and 0.6% vs. 0.8% in the UK). In these countries, the exchange rate appreciation is significant for the inflation-targeting period, but the exchange rate responses are insignificant for the full sample because the exchange rate responses are puzzling in the pre-inflation-

Australia

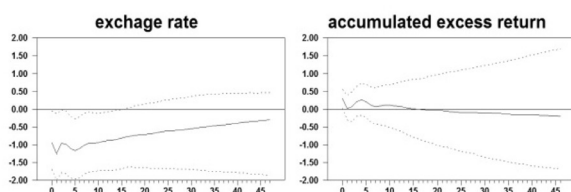
(1) Spread as an Exogenous Variable



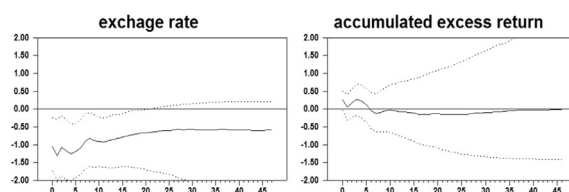
(5) Three Lags for Exogenous Variables



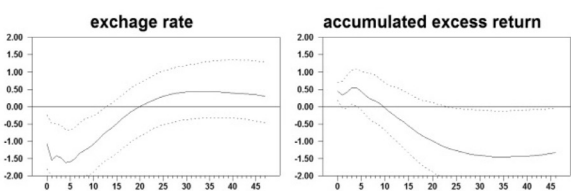
(2) Global Financial Crisis Dummy (08.7–08.12)



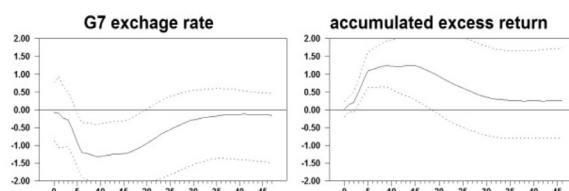
(6) Six Lags for Exogenous Variables



(3) Shadow FFR



(7) G7 Aggregates instead of the US



(4) Non-borrowed Reserves to the Total Reserves as an Exogenous Variable

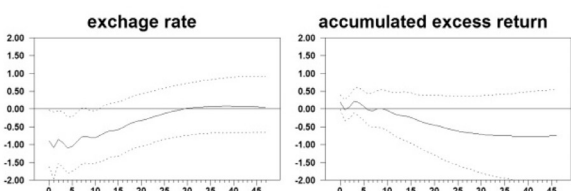


Fig. 9. (continued)

targeting period. This finding suggests that estimating the inflation-targeting period separately for the UK and Australia is important so as to mitigate the puzzling exchange rate responses.

In the pre-inflation-targeting period of Canada, the impact effect on exchange rate is not significant, which differs from that in the inflation-targeting period. In addition, the maximum appreciation is 0.5%, which is smaller than 0.7% as found in the inflation-targeting period. However, the exchange rate appreciation is still significant at some horizons and the maximum effect based on median responses is found only in four months. However, in the full sample period, error bands are huge and exchange rate responses are not significant at any horizons. Exchange rate appreciation is significant in the pre-inflation-targeting and inflation-targeting periods but insignificant in the full sample period combining the two sub-periods. Nevertheless, some differences in the economic structure seem to be present, including monetary policy procedure, in the two sub-periods, so estimating the model for the full sample period does not seem appropriate.

In the full sample period of Sweden, the exchange rate responses tend to be consistent with the theory. The impact appreciation is significant now, and the maximum effect is observed in the second month. The responses tend to be even more consistent with theory than those under the inflation-targeting period.

For the full and pre-inflation-targeting periods, the accumulated risk premium tends to be insignificant as error bands for impulse responses are huge in some cases. However, the accumulated risk premium tends to be still significantly estimated in other cases.

We also compare the contribution of monetary policy shocks to exchange rate fluctuations for the pre-inflation-targeting and full sample periods with that for the inflation-targeting period as shown in Fig. 3. The contribution tends to be larger in the pre-inflation-targeting period than that in the inflation-targeting period for the UK but smaller for Canada and Australia. The contribution tends to be larger in the inflation-targeting period than that in the full sample period for the UK and Sweden. In addition, contribution over time shows different patterns in many cases. For example, the contribution at initial horizons is quite small in the pre-inflation-targeting period for Canada and Australia. Overall, these results suggest the contribution of monetary policy shocks to exchange rate fluctuations in the inflation-targeting period tends to show different patterns than that in the pre-inflation-targeting and full sample periods.

In summary, the results for inflation-targeting period often differ from those for pre-inflation-targeting and full sample periods. In particular, the exchange rate responses are insignificant and/or the dynamic responses are often puzzling for pre-inflation-targeting and/or full sample periods in all countries except for Sweden.^{6,7}

Kim et al. (2017) showed that the delayed overshooting is not observed for the post-Volcker period in the US. Kim et al. (2017) suggested that the imperfect credibility of the Volcker Fed's disinflation policy in the early 1980s is the primary reason delayed overshooting is observed when the Volcker era is included in the sample period. As economic agents considered the Volcker Fed's disinflation to be incredible and expected a policy U-turn, economic agents may underreact to monetary policy shocks, which may generate puzzling responses of exchange rate.

A similar argument may be applied for the three countries in which exchange rate responses are more puzzling when pre-inflation-targeting period is included in the sample period. As discussed in Erceg and Levin (2004), the UK underwent a huge monetary contraction in 1981 to reduce the inflation rate from approximately 17% in the early 1980s to approximately 4% in 1984, and Canada tightened its monetary policy to reduce inflation from approximately 10% in mid-1981 to approximately 3%–4% in 1984. Erceg and Levin (2004) further suggested that inflation forecast errors are large and highly persistent in the UK and Canada during these periods. Australia struggled seriously in stabilizing the inflation rate. In Australia, the transition of monetary policy took a very long time from the early 1980s to early 1990s. Australia struggled with recurring high inflation rate during the 1980s and succeeded in dropping inflation rate sharply only in the late 1980s and early 1990s.

Fig. 7 reports the results when both potential problems are present. That is, the Scholl and Uhlig (2008) model (for a large economy) is used for the sample period starting from the mid-1970s. In general, the exchange rate responses are extremely small and not estimated with any significance in all countries. The size of the median exchange rate responses are smaller than that of the baseline model in all countries. The probability bands for the exchange rate responses include zero responses at all horizons in all countries. The accumulated risk premium tends to be insignificant as error bands for impulse responses are huge.

Overall, these results support the idea that using the sample period for the homogenous monetary policy operating procedure and applying the identification scheme appropriate for small open economies are important to infer correctly the effects of monetary policy shocks on the exchange rates of these small open economies. When such aspects are not considered properly, the exchange rate responses tend to be insignificant and/or involve with a long delay in overshooting.

4. Robustness

In this section, the baseline model is extended in various ways to check the robustness of the results. First, we include the real exchange rate, instead of the nominal exchange rate, to analyze how real exchange rate responds to monetary policy shocks and to measure the contribution of monetary policy shocks on real exchange rate fluctuations. The results are reported in Fig. 8. The results are very similar to those of the baseline model, but the exchange rate responses tend to be slightly weaker than those of the baseline model, which may suggest that the nominal exchange rate and the ratio of the foreign price level to the domestic price level tend to move in opposite directions.

We then consider various additional or replacing exogenous variables. First, the US spread is included as an exogenous variable. This variable represents the worldwide risk condition, which may affect capital flows and exchange rate. Second, we include a dummy variable for the global financial crisis (2008:7–2008:12) because the exchange rate may exhibit abnormal behavior during the global financial crisis. Third, we consider the shadow FFR by Wu and Xia (2015), instead of the FFR. After the global financial crisis, the FFR was close to zero but unconventional monetary policy was implemented. Hence, to consider the unconventional monetary policy of the US properly, we use the shadow rate. Fourth, we include non-borrowed reserves to the total reserves of the US. Some past studies, such as Eichenbaum and Evans (1995), used the variable to identify US monetary policy shocks.

Next, in the baseline model, lags of exogenous variables are not included to save the degree of freedom. Three or six lags of

⁶ The results are similar under tighter restriction, for example, sign restrictions up to 24 months of impulse responses.

⁷ The results for Canada are not much different from those of past studies investigating the sample period, including pre-inflation-targeting period mostly, for example, Cushman and Zha (1997), Kim and Roubini (2001), and Kim (2005), where the delays in overshooting are relatively short. However, the results for the UK are different from those of past studies, such as Kim and Roubini (2001).

exogenous variables are included to check the robustness of the results. Finally, we consider the variables of G7 aggregates, instead of the US variables, as foreign variables in the model. We use the bilateral trade weight to construct the weight for aggregation.

Fig. 9 reports the responses of the exchange rate and risk premium in each case. The results on exchange rate responses are qualitatively similar in most cases, although the error bands are sometimes wide because the degree of freedom decreases. One exception is the case using G7 aggregates for Canada. In Canada, we do not find significant appreciation at any horizon. The effects of the Canadian monetary policy on the exchange rate against the US dollar might be different from the effect on the exchange rate against other G7 countries because the US economy has a very strong influence on Canada, which might explain the result. In other experiments, the main results hold. First, the exchange rate appreciates in response to contractionary monetary policy shocks. The appreciation is significant in most cases. Second, the maximum appreciation is observed within six months after the shock.

Fig. 9 also shows that the estimated deviation of the UIP is relatively small in most cases. The accumulated risk premium for long horizons is not significantly estimated in almost all cases. However, a few cases exist in which the accumulated risk premium for short horizons is significantly estimated, which may reflect a short delay in overshooting of exchange rate.

5. Conclusion

This study empirically investigates the effects of monetary policy shocks on the exchange rates of four small open economies by using VAR models in which sign restrictions are imposed on impulse responses to identify monetary policy shocks. The study focuses on three issues: (1) whether a significant exchange rate appreciation can be found in response to contractionary monetary policy shocks, (2) the length of the delay in overshooting, and (3) whether the deviation from the UIP is significant, conditional on monetary policy shocks. To investigate the issues, we give particular attention to two important considerations. First, we focus on the recent inflation-targeting period during which no substantial changes in monetary policy operating procedure have been observed. Second, we construct the empirical model to reflect properly the features of small open economies.

The empirical results suggest that contractionary monetary policy shocks lead to a significant exchange rate appreciation; that the delay in overshooting is relatively short, at best six months; and that the deviation from the UIP condition is relatively small, although significant at short horizons in some cases. These results clearly different from those of the studies on non-US economies, such as Grilli and Roubini (1995), and US economy, such as Eichenbaum and Evans (1995). In addition, the results are in stark contrast with studies on the US economy of Scholl and Uhlig (2008), who employed VAR models with sign restrictions similar to the current study. Further investigation suggests that considering only the inflation-targeting period (instead of the entire floating exchange rate sample period from the 1970s) and constructing the empirical model that incorporates the features of small open economies are both important to infer correctly the effects of monetary policy shocks on exchange rates.

Appendix

A.1. Data Sources

Variable	Description	Sources and Notes
CR	Money market rate	<i>International Financial Statistics</i>
MB	Monetary base, seasonally adjusted (UK: M1)	Central Bank for each country, except for M1 of the UK from the Federal Reserve Bank of St. Louis Macro Database
CPI	Consumer price index, seasonally adjusted	Federal Reserve Bank of St. Louis, except for CPI of Canada from <i>International Financial Statistics</i> . Interpolation is used for CPI of Australia in which only quarterly data are available.
IP	Industrial production index, seasonally adjusted	<i>International Financial Statistics</i> for the UK and Canada, Federal Reserve Bank of St. Louis Macro Database for Sweden and Australia. Interpolation is used for IP of Australia in which only quarterly data are available.
ERA	Exchange rate of domestic currency per US dollars (period average)	<i>International Financial Statistics</i> , except for ERA of the UK from the Federal Reserve Bank of St. Louis Macro Database.
FR	Foreign exchange reserve (total reserves minus gold)	<i>International Financial Statistics</i>
FFR	Federal funds rate	<i>International Financial Statistics</i>
USCPI	US consumer price index, seasonally adjusted	<i>International Financial Statistics</i>
USIP	US industrial production index, seasonally adjusted	<i>International Financial Statistics</i>
SPREAD	Moody's seasoned Baa corporate – 10-year US Treasury constant maturity rate	Federal Reserve Bank of St. Louis Macro Database

A.2. Structural VAR Models with Sign Restrictions

The methodology of the structural VAR model with sign restrictions is described briefly. The following reduced form of the VAR model is considered:

$$Y_t = B(L)Y_{t-1} + C(L)X_t + u_t, \quad (\text{A.1})$$

where Y_t is an $l \times 1$ vector of endogenous variables, X_t is an $m \times 1$ vector of exogenous variables, u_t is a $l \times 1$ residual vector, $E(u_t) = 0$, $E(u_t u_t') = \Sigma$, and $B(L)$ and $C(L)$ are $l \times l$ and $l \times m$ matrix polynomials in lag operator L .

In previous studies, reduced-form residuals and elements of u_t are the linear combinations of structural shocks:

$$u_t = A v_t, \quad (\text{A.2})$$

where A is an $l \times l$ matrix, v_t is an $l \times 1$ vector of structural shocks, $E(v_t) = 0$, and $E(v_t v_t') = 1$. Previous studies often recovered orthogonal structural shocks from reduced-form residuals by determining A . For example, the recursive identification strategy developed by Sims (1980) recovers A as a lower triangular matrix by applying Cholesky decomposition on Σ .

Uhlig (2005) identified structural shocks by imposing sign restrictions on impulse responses. The study identified only one structural shock in particular, that is, monetary policy shocks, which amounts to identifying a single column $a \in R^m$ of the matrix A . Uhlig (2005) defined the impulse vector as follows.

Definition 1. The vector $a \in R^m$ is called an impulse vector if matrix A exists; thus, $AA' = \Sigma$ and a is a column of A .

Uhlig (2005) showed that any impulse vector a can be characterized by $a = \tilde{A}\alpha$, where $\tilde{A}\tilde{A}' = \Sigma$ is a Cholesky decomposition of Σ and α is an l -dimensional vector of unit length. Thereafter, the vector impulse response $r_a(k)$ for a is expressed as $r_a(k) = \sum_{j=1}^l \alpha_j r_j(k)$, where $r_j(k) \in R^l$ is the vector response at horizon k to the j th shock in a Cholesky decomposition of Σ . A list of inequality restrictions on the entries of the vector impulse response $r_a(k)$ at various horizons k is then imposed.

Following the pure sign restriction approach by Uhlig (2005), a Bayesian prior for the VAR parameters (B , Σ) and an independent uniform prior for α are assumed. Only the draws that satisfy the inequality restrictions are retained in the simulation exercise. The probability bands are calculated based on 5,000 draws.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jmacro.2018.04.008](https://doi.org/10.1016/j.jmacro.2018.04.008).

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